

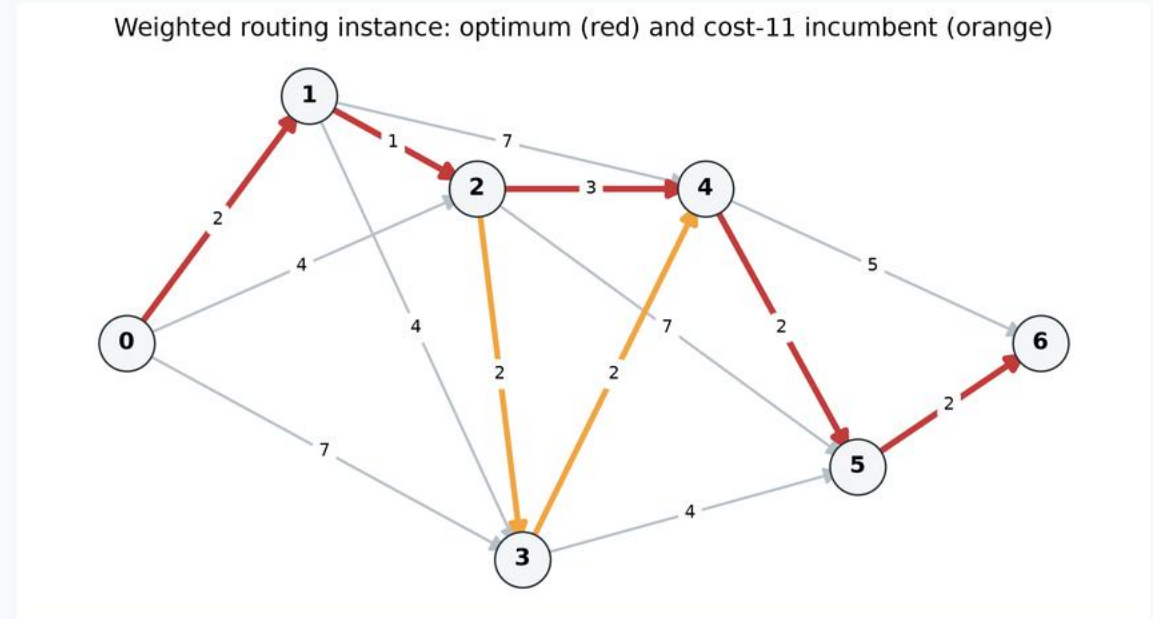
QAOA Routing

From a weighted graph to QUBO, Ising, QAOA, and decoded routes

Ideal statevector simulation · teaching-scale instance · no quantum-advantage claim

The routing problem

- 7 nodes and 14 directed weighted edges
- Source 0, target 6; one binary variable per edge
- Unique optimum: $0 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 6$
- Exact route cost: 10



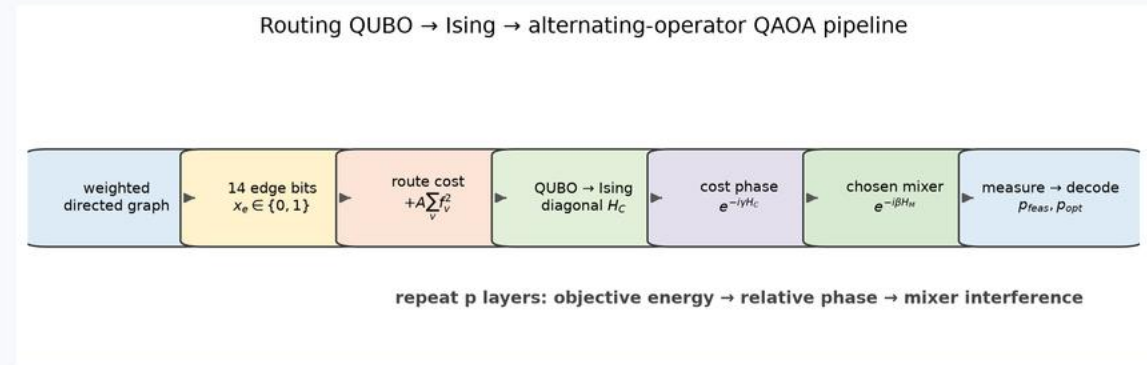
Edge variables become a routing QUBO

- $x_e = 1$ means that directed edge e is selected
- Routing cost is the weighted sum of selected edges
- Squared flow residuals enforce source, conservation, and target constraints
- The frozen penalty coefficient is $A = 6$

$$Q_A(x) = \sum_e w_e x_e + A P_{\text{flow}}(x)$$

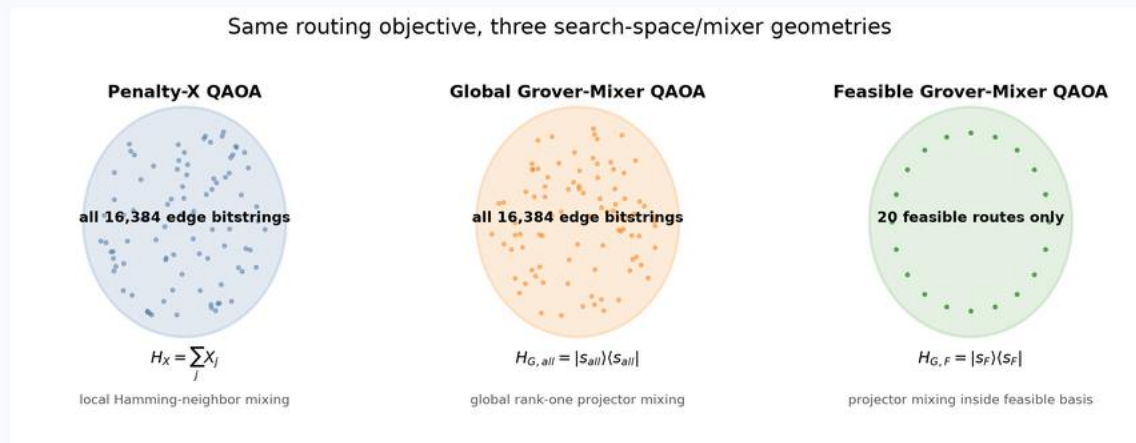
QUBO $\rightarrow$ Ising $\rightarrow$ QAOA

- Substitute $x_j = (1 - Z_j) / 2$
- The diagonal cost Hamiltonian contains I , Z , and ZZ terms
- A cost phase changes phases; the mixer converts phase differences into probability changes
- All 16,384 QUBO and Ising basis energies agree exactly



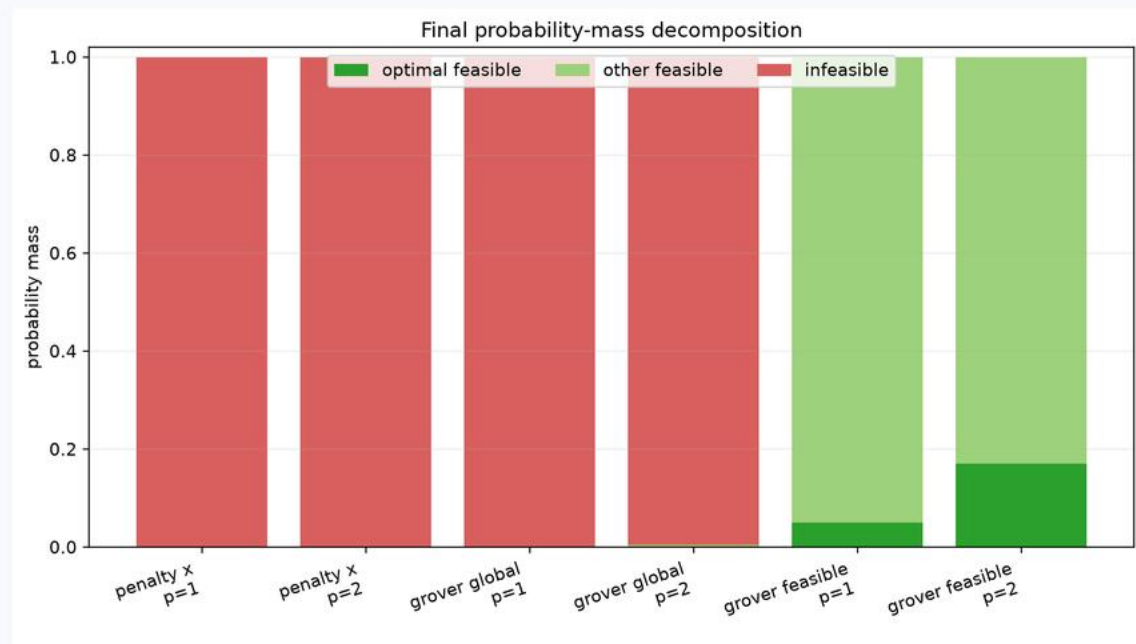
Two search spaces, three mixers

- Penalty-X: $2^{14} = 16,384$ edge bitstrings
- Global-Grover: the same full bitstring space
- Feasible-Grover: only 20 enumerated valid routes
- Global-Grover and Feasible-Grover are QAOA mixers, not standard Grover search



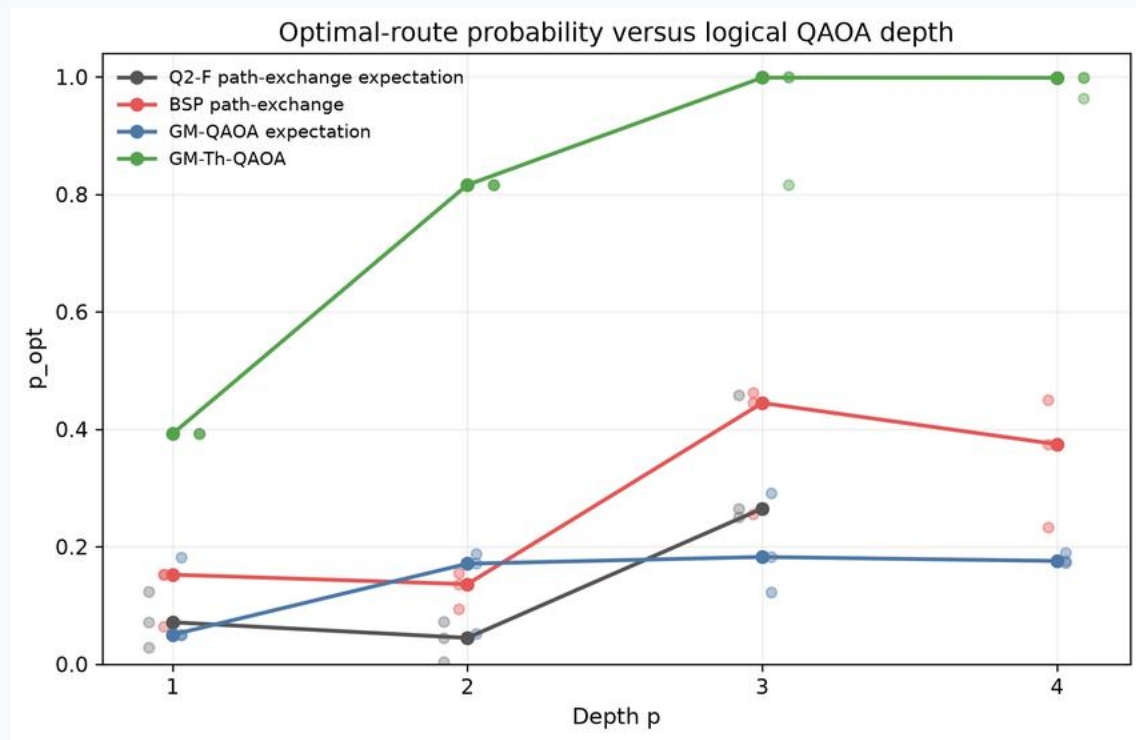
What happens to probability mass?

- Full-space initial $p_{\text{feas}} = 20 / 16,384 \approx 0.00122$
- At $p=2$, Global-Grover reaches $p_{\text{opt}} \approx 0.000213$
- At $p=2$, Feasible-Grover keeps $p_{\text{feas}} = 1$ and reaches $p_{\text{opt}} \approx 0.170927$
- Feasible restriction solves validity; phase and mixer still determine optimality



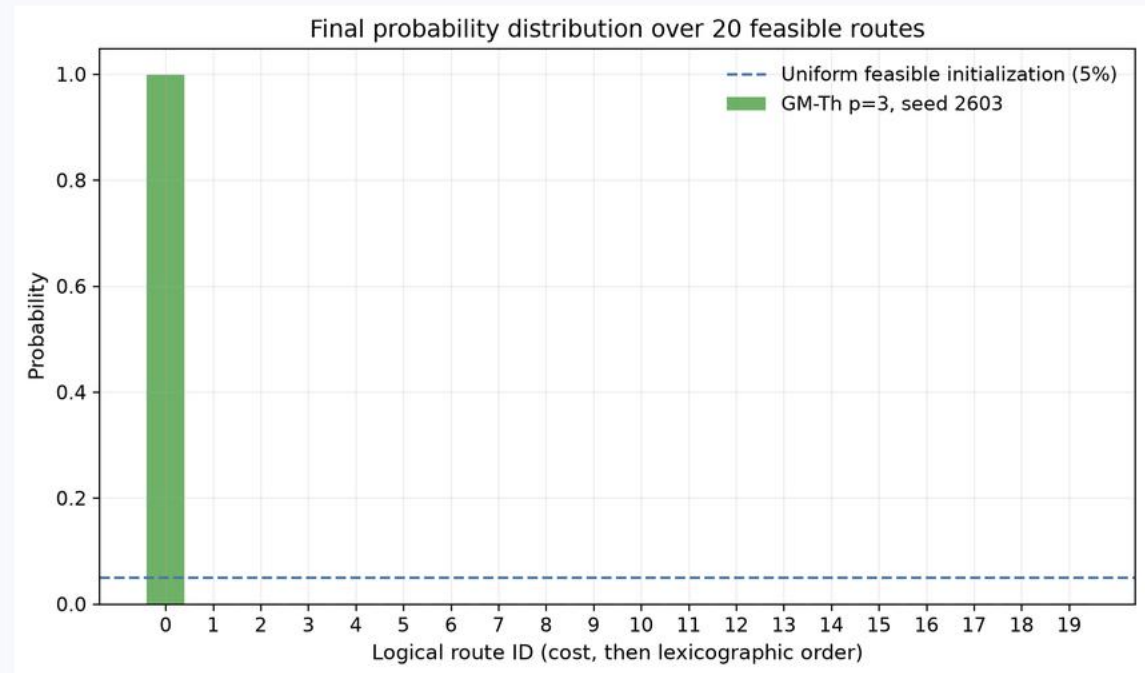
Final course method comparison

- Q2-F Expectation, $p=3$: 0.264901
- BSP / path exchange, $p=3$: 0.444656
- GM-QAOA Expectation, $p=3$: 0.182410
- GM-Th-QAOA, $p=3$: 0.998712
- All four methods use the 20-route feasible basis, so $p_{\text{feas}} \approx 1$



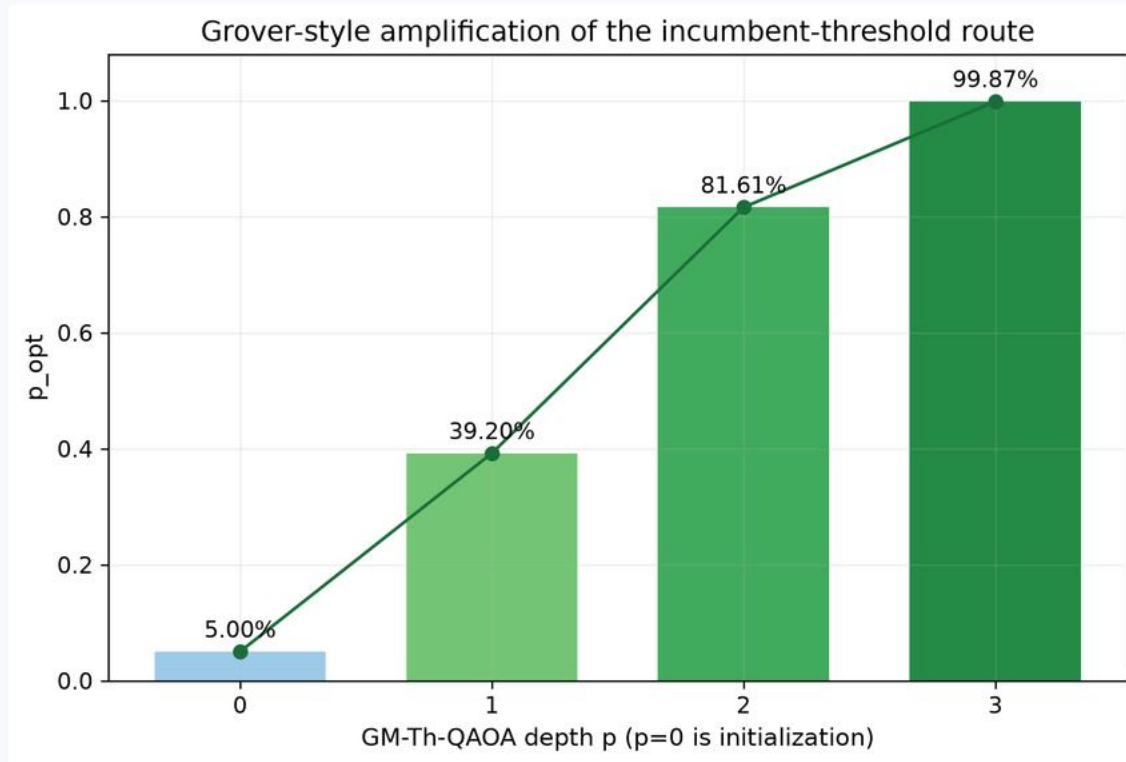
The final distribution is concentrated on the optimum

- Median $p_{\text{opt}} = 0.998712$
- Seed range: 0.816034–0.999804
- Median expected route cost = 10.002983
- The highlighted state is the unique cost-10 route



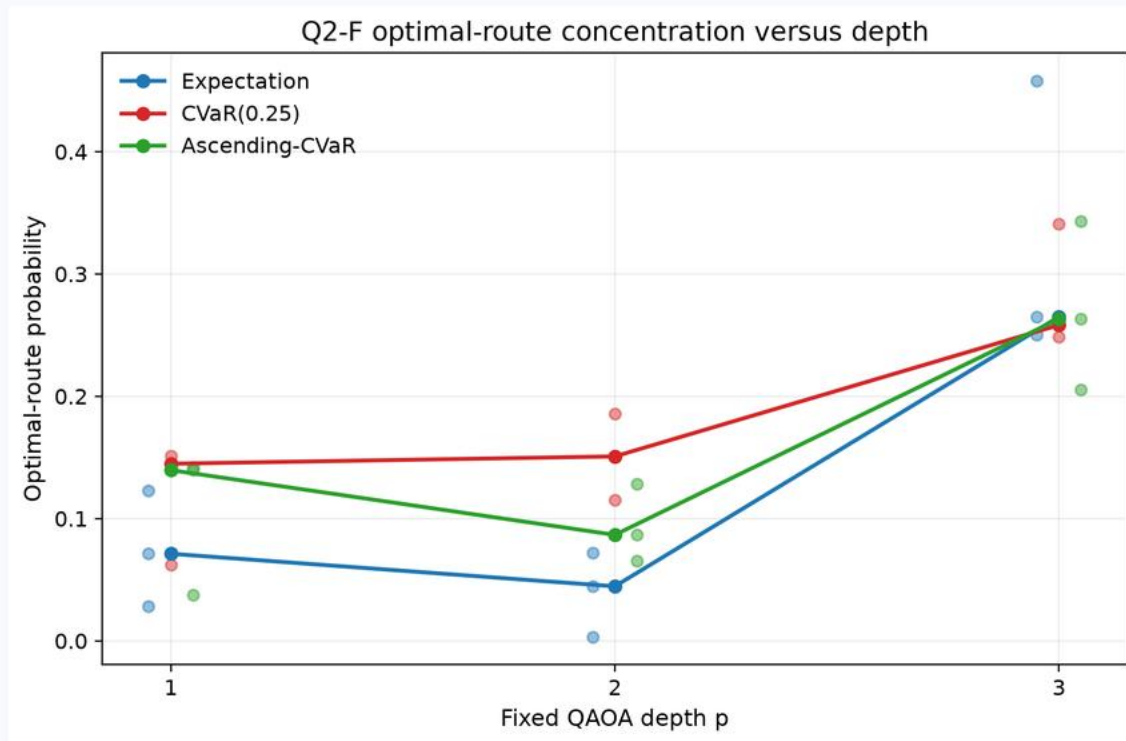
Why GM-Th works on this instance

- The logical basis contains only valid routes: $p_{\text{feas}} = 1$
- A deterministic greedy incumbent has cost 11
- The threshold marks routes with cost < 11
- On this graph, that marked set contains exactly the unique optimum
- The feasible Grover mixer amplifies the marked probability



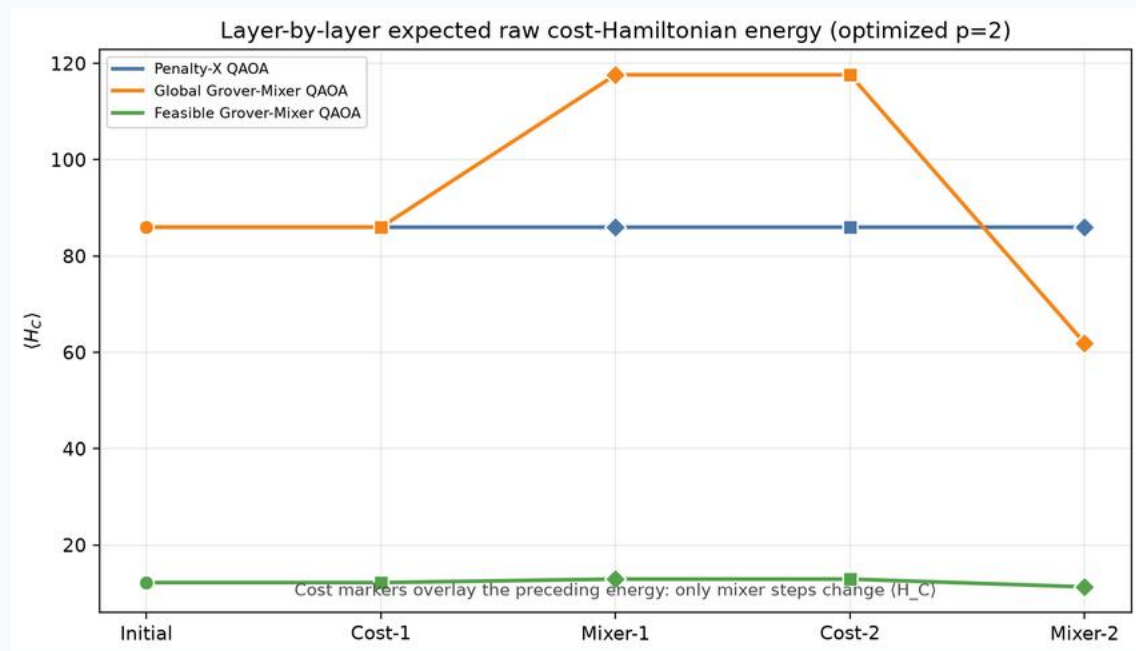
CVaR course extension: a useful negative result

- CVaR changes the classical objective, not the mixer
- At depth 3: Expectation 0.264901, CVaR 0.258176, Ascending CVaR 0.263190
- CVaR did not provide a clear improvement in this course experiment
- The fractional cutoff implementation remains tested and available



Run it and watch it

- `python -m src.main`
- `python scripts/run_course_final.py --seed 2601 --depth 1`
- `python scripts/run_qaoa_visualizer.py`
- `python scripts/run_qaoa_dynamics_visualizer.py`
- Both browser demos replay saved data and do not rerun historical experiments



What we learned

- Graph $\rightarrow$ QUBO $\rightarrow$ Ising $\rightarrow$ QAOA $\rightarrow$ decoded routes works end to end
- Full-space QAOA can waste most probability on infeasible bitstrings
- Feasibility preservation does not automatically imply optimal-state concentration
- Mixer and phase/objective design strongly change the final distribution
- One fixed graph, ideal simulation, enumerated feasible basis, finite optimizer budgets
- No hardware, scalability, or quantum-advantage claim

Backup: a two-qubit Ising gate example

- Local Z terms become RZ rotations
- A ZZ coupling becomes CX-RZ-CX
- The X mixer becomes RX rotations
- This is a small teaching decomposition, not the full 14-qubit compiled circuit

